1.9.5

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Question

Find the distance between the points $(0, 2\sqrt{5})$ and $(-2\sqrt{5},\ 0)$.

Theoretical Solution

Let the points be $\mathbf{A} = (0, 2\sqrt{5})$ and $\mathbf{B} = (-2\sqrt{5}, 0)$.

$$\mathbf{A} = \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \tag{1}$$

The distance between two points is given by

$$d = \|\mathbf{A} - \mathbf{B}\| \tag{2}$$

Substituting values,

$$d = \left\| \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix} - \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \right\| \tag{3}$$

$$= \left\| \begin{pmatrix} 2\sqrt{5} \\ 2\sqrt{5} \end{pmatrix} \right\| \tag{4}$$

$$= \sqrt{(2\sqrt{5})^2 + (2\sqrt{5})^2} \tag{5}$$

Simplifying,

$$d = \sqrt{20 + 20} \tag{6}$$

$$= \sqrt{40} \tag{7}$$

$$= 2\sqrt{10} \tag{8}$$

$$\implies d = \boxed{2\sqrt{10}} \tag{9}$$

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

// Function to compute distance between (x1,y1) and (x2,y2)
double find_distance(double x1, double y1, double x2, double y2)
    {
    double dx = x2 - x1;
    double dy = y2 - y1;
    return sqrt(dx*dx + dy*dy);
}

int main() {
    double x1 = 0, y1 = 2*sqrt(5);
    double x2 = -2*sqrt(5), y2 = 0;
    double d = find_distance(x1, y1, x2, y2);
    printf(Distance = %.4f\n, d);
    return 0;
}
```

Python + C Code

```python
import ctypes
import numpy as np
import matplotlib.pyplot as plt
import math

# Load shared library
lib = ctypes.CDLL('./section.so')

# Define function signature
lib.find_distance.argtypes = [ctypes.c_double, ctypes.c_double,
                              ctypes.c_double, ctypes.c_double]
lib.find_distance.restype = ctypes.c_double

# Points
x1, y1 = 0, 2*math.sqrt(5)
x2, y2 = -2*math.sqrt(5), 0

# Call C function
dist = lib.find_distance(x1, y1, x2, y2)
```